

BANK DATABASE – ACID PROPERTIES & ISOLATION

2520030021

Karuturi Sai Akhil

Create Database

```
1 • CREATE DATABASE Bank_ACID_DB;  
2 • USE Bank_ACID_DB;
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	19:24:31	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
✓ 2	19:24:37	USE Bank_ACID_DB	0 row(s) affected

Create Customer Table , Accounts Table and Bank Transaction Table

```
10     Account_No INT PRIMARY KEY,
11     Customer_ID INT,
12     Account_Type VARCHAR(20),
13     Balance DECIMAL(12,2),
14     Branch VARCHAR(50),
15
16     FOREIGN KEY (Customer_ID)
17     REFERENCES Customer(Customer_ID)
18 );
19 • CREATE TABLE Bank_Transaction (
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21     Account_No INT,
22     Transaction_Type VARCHAR(20),
23     Amount DECIMAL(12,2),
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25
26     FOREIGN KEY (Account_No)
27     REFERENCES Account(Account_No)
28 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	19:24:31	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
✓ 2	19:24:37	USE Bank_ACID_DB	0 row(s) affected
✓ 3	19:25:48	CREATE TABLE Customer (Customer_ID I...	0 row(s) affected
✓ 4	19:25:58	CREATE TABLE Account (Account_No IN...	0 row(s) affected
✓ 5	19:26:04	CREATE TABLE Bank_Transaction (Trans...	0 row(s) affected

Insert Customer Data,Insert Account Data ,Insert Bank Transaction Data

SQL File 11*

```

31 VALUES
32 (101, 'Sai Akhil', '2520030021', 'Hyderabad'),
33 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
34 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
35 (104, 'Sneha Rao', '9876543213', 'Chennai'),
36 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
37 • INSERT INTO Account
38 (Account_No, Customer_ID, Account_Type, Balance, Branch)
39 VALUES
40 (10001, 101, 'Savings', 50000, 'Hyderabad'),
41 (10002, 102, 'Savings', 75000, 'Vijayawada'),
42 (10003, 103, 'Current', 120000, 'Bangalore'),
43 (10004, 104, 'Savings', 45000, 'Chennai'),
44 (10005, 105, 'Current', 90000, 'Hyderabad');
45 • INSERT INTO Bank_Transaction
46 (Account_No, Transaction_Type, Amount)
47 VALUES
48 (10001, 'DEPOSIT', 10000),
49 (10002, 'DEPOSIT', 15000),
50 (10003, 'WITHDRAWAL', 20000)

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 3	19:25:48	CREATE TABLE Customer (Customer_ID ...	0 row(s) affected	0.047 sec
✓ 4	19:25:58	CREATE TABLE Account (Account_No I...	0 row(s) affected	0.047 sec
✓ 5	19:26:04	CREATE TABLE Bank_Transaction (Tra...	0 row(s) affected	0.062 sec
✓ 6	19:27:35	INSERT INTO Customer (Customer_ID, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.000 sec
✓ 7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.015 sec
✓ 8	19:27:54	INSERT INTO Bank_Transaction (Account_...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.016 sec

Display Data

53 • `SELECT * FROM Customer;`

Result Grid | Filter Rows: | Edit: | Export/Import:

Customer_ID	Customer_Name	Phone	City
101	Sai Akhil	2520030021	Hyderabad
102	Priya Sharma	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	Sneha Rao	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad
NULL	NULL	NULL	NULL

Customer 1 x

Output

Action Output

#	Time	Action	Message
4	19:25:58	CREATE TABLE Account (Account_No...	0 row(s) affected
5	19:26:04	CREATE TABLE Bank_Transaction (Tr...	0 row(s) affected
6	19:27:35	INSERT INTO Customer (Customer_ID, Cu...	5 row(s) affected Records: 5 Duplicates: 0
7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0
8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicates: 0
9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

55 • `SELECT * FROM Account;`

Result Grid   Filter Rows: Edit:    Export/Import:   W

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	Current 0	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 2 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	5	19:26:04	CREATE TABLE Bank_Transaction (Tr...	0 row(s) affected
✓	6	19:27:35	INSERT INTO Customer (Customer_ID, Cu...	5 row(s) affected Records: 5 Duplicates: 0 ...
✓	7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...
✓	8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicates: 0 ...
✓	9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓	10	19:28:53	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

56

57 • `SELECT * FROM Bank_Transaction;`

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 19:27:54
2	10002	DEPOSIT	15000.00	2026-08-15 19:27:54
3	10003	WITHDRAW	20000.00	2026-08-15 19:27:54
4	10004	DEPOSIT	5000.00	2026-08-15 19:27:54
5	10005	WITHDRAW	10000.00	2026-08-15 19:27:54
NULL	NULL	NULL	NULL	NULL

Bank_Transaction 3 x

Apply

Output

Action Output

#	Time	Action	Message	Duration
6	19:27:35	INSERT INTO Customer (Customer_ID, Cu...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.000 se
7	19:27:45	INSERT INTO Account (Account_No, Cust...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.015 se
8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicates: 0 ...	0.016 se
9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 se
10	19:28:53	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 se
11	19:29:12	SELECT * FROM Bank_Transaction LIMIT ...	5 row(s) returned	0.000 se

Start Transaction

```

58 • START TRANSACTION;
59 • UPDATE Account
60   SET Balance = Balance + 5000
61   WHERE Account_No = 10001;

```

Output



Action Output

	#	Time	Action	Message
✓	8	19:27:54	INSERT INTO Bank_Transaction (Account...	5 row(s) affected Records: 5 Duplicate
✓	9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓	10	19:28:53	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓	11	19:29:12	SELECT * FROM Bank_Transaction LIMIT ...	5 row(s) returned
✓	12	19:31:40	START TRANSACTION	0 row(s) affected
✓	13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Cl

Check Balance


```

58 • START TRANSACTION;
59 • UPDATE Account
60   SET Balance = Balance + 5000
61   WHERE Account_No = 10001;
62 • SELECT *
63   FROM Account
64   WHERE Account_No = 10001;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	W
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 4 x

Output

Action Output

#	Time	Action	Message
✓ 9	19:28:38	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	19:28:53	SELECT * FROM Account	SELECT * FROM Account returned
✓ 11	19:29:12	SELECT * FROM Bank_Tr LIMIT 0, 1000	returned
✓ 12	19:31:40	START TRANSACTION	0 row(s) affected
✓ 13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...
✓ 14	19:32:23	SELECT * FROM Account WHERE Accou...	1 row(s) returned

Commit and Check Transaction After commit


```

62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
66 • SELECT *
67 FROM Account
68 WHERE Account_No = 10001;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap C

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 5 x Ap

Output

Action Output

	#	Time	Action	Message	Dur
✓	11	19:29:12	SELECT * FROM Bank_Transaction LIMIT ...	5 row(s) returned	0.00
✓	12	19:31:40	START TRANSACTION	0 row(s) affected	0.00
✓	13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...	0.00
✓	14	19:32:23	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.00
✓	15	19:32:41	COMMIT	0 row(s) affected	0.00
✓	16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.00

Start Transaction and Withdraw 10,000

```

68     WHERE Account_No = 10001;
69 •   START TRANSACTION;
70 •   UPDATE Account
71     SET Balance = Balance - 10000
72     WHERE Account_No = 10001;

```

Output:

Action Output				
	#	Time	Action	Message
✓	13	19:31:47	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...
✓	14	19:32:23	SELECT * FROM Account WHERE Accou...	1 row(s) returned
✓	15	19:32:41	COMMIT	0 row(s) affected
✓	16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned
✓	17	19:34:15	START TRANSACTION	0 row(s) affected
✓	18	19:34:22	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...

Check Balance

Limit to 1000 rows

```

69 • START TRANSACTION;
70 • UPDATE Account
71   SET Balance = Balance - 10000
72   WHERE Account_No = 10001;
73 • SELECT *
74   FROM Account
75   WHERE Account_No = 10001;

```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 6 x Apply

Output

Action Output

#	Time	Action	Message	Duration / Fe
✓ 14	19:32:23	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.000 sec / (
✓ 15	19:32:41	COMMIT	0 row(s) affected	0.000 sec
✓ 16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.000 sec / (
✓ 17	19:34:15	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 18	19:34:22	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec
✓ 19	19:34:39	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.000 sec / (

RollBack and Check Balance after rollback

```

76 • ROLLBACK;
77 • SELECT *
78 FROM Account
79 WHERE Account_No = 10001;

```

Result Grid					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 7 ×					
Apply					
Output					
Action Output					
#	Time	Action	Message	Duration / Fe	
✓ 15	19:32:41	COMMIT	0 row(s) affected	0.000 sec	
✓ 16	19:32:48	SELECT * FROM Account WHERE Accou...	1 row(s) returned	0.000 sec /	
✓ 17	19:34:15	START TRANSACTION	0 row(s) affected	0.000 sec	
✓ 18	19:34:22	UPDATE Account SET Balance = Balance...	1 row(s) affected Rows matched: 1 Chang...	0.015 sec	
✓ 19	19:34:39	SELECT * FROM A	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	returned	0.000 sec /
✓ 20	19:35:15	SELECT * FROM A	returned	0.000 sec /	